

4.4 Chebyshev's Theorem

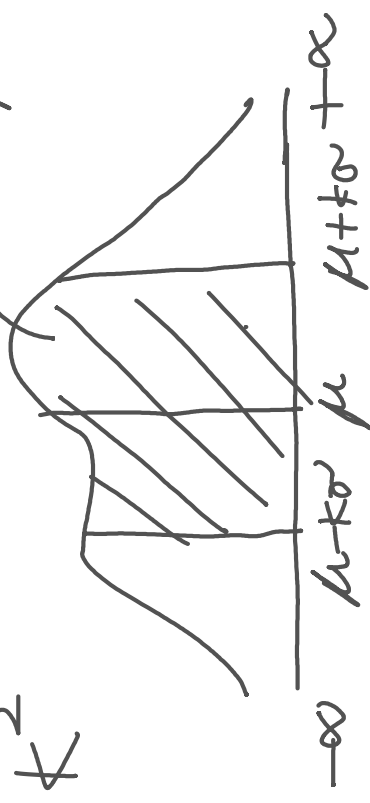
Theorem 4.8. (Chebyshev's Theorem)

If μ and σ are the mean and the standard deviation of a random variable X , then for any positive constant k , then probability is at least $(1 - 1/k^2)$ that X will take on a value within k standard deviation of the mean, symbolically,

Lower bound for P .

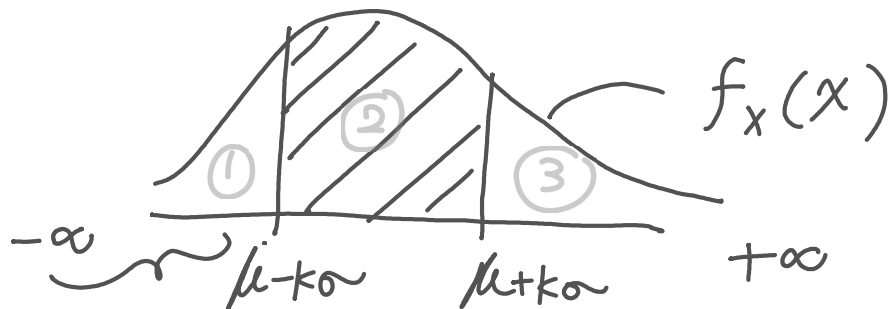
$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

$$\mu - k\sigma < X < \mu + k\sigma$$



$$|X| < 5 \Rightarrow -5 < X < +5$$
$$|X - \mu| < k\sigma$$

Proof



$$\sigma^2 = E[(X-\mu)^2] = \int_{-\infty}^{+\infty} (x-\mu)^2 f_X(x) dx$$

$$= \int_{-\infty}^{\mu-k\sigma} (x-\mu)^2 f_X(x) dx + \int_{\mu-k\sigma}^{\mu+k\sigma} (x-\mu)^2 f_X(x) dx + \int_{\mu+k\sigma}^{+\infty} (x-\mu)^2 f_X(x) dx$$

$$\sigma^2 \geq \int_{-\infty}^{\mu-k\sigma} (x-\mu)^2 f_X(x) dx + \int_{\mu+k\sigma}^{+\infty} (x-\mu)^2 f_X(x) dx$$

Region I: $x < \mu - k\sigma$

Region II: $x > \mu + k\sigma$

$$x < \mu - k\sigma$$

$$x > \mu + k\sigma$$

$$x - \mu < -k\sigma$$

$$x - \mu > k\sigma$$

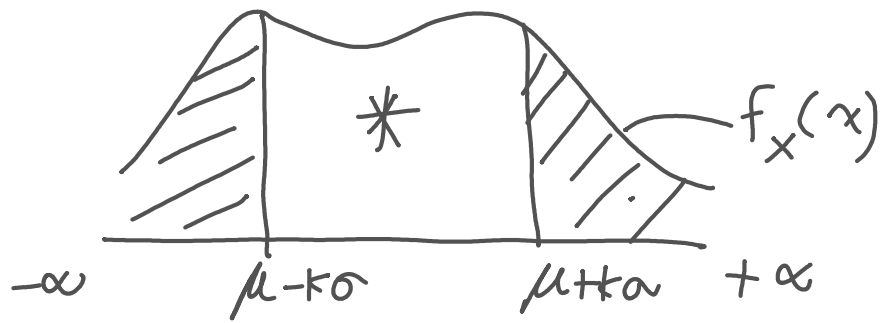
$$(x-\mu)^2 > k^2 \sigma^2$$

$$(x-\mu)^2 > k^2 \sigma^2$$

$$\Rightarrow \sigma^2 \geq \int_{-\infty}^{\mu-k\sigma} k^2 \sigma^2 f_X(x) dx + \int_{\mu+k\sigma}^{+\infty} k^2 \sigma^2 f_X(x) dx$$

$$\frac{1}{k^2} \geq \int_{-\infty}^{\mu-k\sigma} f_X(x) dx + \int_{\mu+k\sigma}^{+\infty} f_X(x) dx$$

$$\frac{1}{k^2} \geq \int_{-\infty}^{\mu-k\sigma} f_X(x) dx + \int_{\mu+k\sigma}^{+\infty} f_X(x) dx$$



$$\frac{1}{k^2} \geq P(X < \mu - k\sigma) + P(X > \mu + k\sigma)$$

$$P(X < \mu - k\sigma) + P(X > \mu + k\sigma) \leq \frac{1}{k^2}$$

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}$$

$$P(|X - \mu| \leq k\sigma) \geq 1 - \frac{1}{k^2}$$

$$-k\sigma < x - \mu < +k\sigma$$

$$\mu - k\sigma < x < \mu + k\sigma$$

lower bound of

Example. Find probabilities for 2σ , 3σ , 4σ

$$P(|x - \mu| < \underset{\uparrow k}{2}\sigma) \geq 1 - \frac{1}{k^2} = 1 - \frac{1}{2^2} = 1 - \frac{1}{4} = 75\%$$

$$P(|x - \mu| < 3\sigma) \geq 1 - \frac{1}{3^2} = 1 - \frac{1}{9} = \frac{8}{9} = 88.8\%$$

Note: The probability given by Chebyshev's Theorem is only a lower bound . In other words it assumes that this probability cannot be less than $1 - 1/k^2$. Only when the distribution of a random variable is known, we calculate the exact probability